

Active Directory Homelab Project

Overview

This project involved deploying and configuring a Windows Server 2022 Active Directory environment within Oracle VirtualBox. The purpose of the lab was to simulate a small enterprise network environment and develop practical hands-on experience with Windows infrastructure, identity management, and systems administration.

The environment included:

- Windows Server 2022 Domain Controller
- Active Directory Domain Services (AD DS)
- Organisational Unit (OU) structure
- Domain user and security group management
- Windows 11 domain-joined workstation
- Group Policy Management
- PowerShell-based Active Directory enumeration

This project was completed as part of my cybersecurity and infrastructure portfolio to strengthen practical enterprise IT administration skills relevant to IT support, infrastructure, and SOC analyst roles.

Lab Environment

Component	Description
Hypervisor	Oracle VirtualBox
Server Operating System	Windows Server 2022 Standard Evaluation
Client Operating System	Windows 11
Domain Name	pavlab.local
Domain Controller	DC01

Objectives

The objectives of this project were to:

- Deploy a Windows Server 2022 virtual machine
 - Install and configure Active Directory Domain Services (AD DS)
 - Promote the server to a Domain Controller
 - Create a new Active Directory forest and domain
 - Configure Organisational Units (OUs)
 - Create and manage domain users and security groups
 - Perform password resets and account administration tasks
 - Join a Windows 11 client machine to the domain
 - Verify domain objects using PowerShell
 - Access and verify Group Policy Management functionality
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Active Directory Configuration

Domain Configuration

Setting	Value
Domain Name	pavlab.local
NetBIOS Name	PAVLAB
Server Name	DC01

The domain controller was configured to provide:

- Authentication services
 - Directory services
 - Centralised identity management
 - Domain infrastructure services
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Organisational Unit Structure

The following Organisational Units (OUs) were created:

- HR
- IT
- Workstations
- Disabled Users

This structure demonstrates administrative separation and allows future Group Policy targeting.

User Account Administration

User Accounts Created

Name	Username	Status
John JS. Smith	jsmith	Enabled
Martin MB. Brown	mbrown	Disabled

Tasks completed included:

- Creating user logon names
 - Configuring User Principal Names (UPN)
 - Managing password settings
 - Reviewing account properties
 - Disabling user accounts
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Security Group Management

A security group named **IT-Support** was created to demonstrate role-based access control.

Tasks performed:

- Created the IT-Support security group
- Added users to the group
- Verified group membership
- Reviewed group member relationships

This demonstrates how Active Directory security groups simplify permissions management within enterprise environments.

Password and Account Management

Administrative tasks performed:

- Reset user passwords
- Forced password change at next logon
- Disabled user accounts
- Verified disabled account status

These are common administrative tasks performed by IT support and systems administration teams.

Windows 11 Domain Join

A Windows 11 client machine was successfully joined to the `pavlab.local` domain.

After joining the domain:

- The workstation object appeared inside Active Directory
- The device was verified inside the Workstations OU
- Domain communication was successfully established

This demonstrates centralised workstation management within a domain environment

Group Policy Management

The Group Policy Management Console (GPMC) was accessed and verified following Active Directory deployment.

The Group Policy infrastructure allows administrators to:

- Enforce security policies
 - Configure workstation settings
 - Apply centralised administrative controls
 - Standardise enterprise configurations
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PowerShell Enumeration

PowerShell was used to enumerate and verify Active Directory objects and domain information.

Tasks performed:

- Retrieved Active Directory domain details
- Verified domain configuration information
- Enumerated user account information
- Reviewed enabled and disabled account states

This demonstrates the use of PowerShell for scalable administration and automation within enterprise Windows environments.

Technologies Used

- Windows Server 2022
 - Active Directory Domain Services (AD DS)
 - Active Directory Users and Computers (ADUC)
 - Group Policy Management Console (GPMC)
 - Windows PowerShell
 - Oracle VirtualBox
 - Windows 11
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Skills Demonstrated

This project demonstrates practical experience in:

- Active Directory administration
 - Windows Server deployment
 - Domain Controller configuration
 - Organisational Unit management
 - User and group administration
 - Identity and Access Management (IAM)
 - Password and account life-cycle management
 - Group Policy Management
 - Windows workstation domain integration
 - PowerShell administration
 - Virtualised lab deployment
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Conclusion

This project successfully demonstrated the deployment and administration of a Windows Active Directory environment using Windows Server 2022 and Oracle VirtualBox.

Through this lab, practical experience was gained in:

- Enterprise identity management
- User and group administration
- Domain infrastructure deployment
- Centralised workstation management
- Active Directory administrative tools
- PowerShell-based systems administration

This project provided hands-on exposure to technologies and administrative tasks commonly performed within enterprise IT support, infrastructure, and cybersecurity environments.